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- Predictive Performance
- Measuring Performance
- Test and training error
- Estimating the test error
- Bias and Variance
- Regularisation
- Cross-validation

Machine learning – Block 1(b)

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Section 1

Predictive Performance



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Previous Model Evaluation

- In the past, tools for assessing models, e.g.:
  - Residuals
  - Leverage, Cook's distance
  - p-values
  - R^2
  - AIC
  - (LOO-CV)
- Model diagnoses and how well the model fits the data.



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- Statistics: estimation and attribution



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- Model diagnoses and how well the model fits the data.
- Statistics: estimation and attribution
- Supervised learning: predictive performance



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Predictive Performance

How well our model $\hat{f}_{\mathcal{T}}$ trained on

$$\mathcal{T} = \{(y_i, x_i), i = 1, ..., n\}$$

work when predicting a new observation (y_0, x_0) from the data generating process $P_{y,x}$.

$$\mathbb{E}\left[L(y_0, \hat{f}_{\mathcal{T}}(x_0))\right] = \int L(y_0, \hat{f}_{\mathcal{T}}(x_0)) P_{(y,x)} d(y_0, x_0)$$

where $L(y, x)$ is a loss function (e.g. $L(x, y) = (x - y)^2$)



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Predictive Performance

- The ability to perform well on previously unobserved inputs is called generalization
- $\mathbb{E}\left[L(y_0, \hat{f}_{\mathcal{T}}(x_0))\right]$ is the generalization error



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Predictive Performance

- The ability to perform well on previously unobserved inputs is called generalization
- $\mathbb{E}\left[L(y_0, \hat{f}_{\mathcal{T}}(x_0))\right]$ is the generalization error
- Models can overfit
  - explain training data well
  - poor generalizability



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Predictive Performance

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- Models can underfit
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Section 2

Measuring Performance



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Loss Functions

- To assess the performance we use the loss function for a new unseen observation y_0 and the prediction of that observation $\hat{f}(x_0)$

$$L(y_0, \hat{f}(x_0))$$



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- Choose L according to what we want the model to perform well on.



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- Examples:
  - Regression problems (squared loss/error):

$$L(y_0, \hat{f}(x_0)) = (y_0 - \hat{f}(x_0))^2$$



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$$L(y_0, \hat{f}(x_0)) = (y_0 - \hat{f}(x_0))^2$$

  - Classification (0-1 loss)

$$L(y_0, \hat{f}(x_0)) = I(y_0 \neq \hat{f}(x_0))$$



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- Examples:
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$$L(y_0, \hat{f}(x_0)) = I(y_0 \neq \hat{f}(x_0))$$

  - The negative log likelihood is a good general loss function



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Cross-Entropy Loss

- When we predict probabilities $\hat{f}(x_0) = \hat{p}$:

$$L(y_0, \hat{p}) = -[(y_0 \log \hat{p}) + (1 - y_0) \log (1 - \hat{p})]$$

Question: Do you recognize the (cross-entropy) loss function?



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  Question: Do you recognize the (cross-entropy) loss function?

- Maximizing the likelihood is the same as minimizing the cross-entropy.



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Cross-Entropy Loss

- When we predict probabilities $\hat{f}(x_0) = \hat{p}$:

$$L(y_0, \hat{p}) = -[(y_0 \log \hat{p}) + (1 - y_0) \log (1 - \hat{p})]$$

  Question: Do you recognize the (cross-entropy) loss function?

- Maximizing the likelihood is the same as minimizing the cross-entropy.
- Multi class cross-entropy over M classes

$$L(\mathbf{y}_0, \hat{\mathbf{p}}) = -\sum_{j=1}^{M} y_{0,j} \log \hat{p}_j$$



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The Confusion Matrix

- A common way to present performance in classification is the confusion matrix:

	Prediction	
Actual	Positive	Negative
Positive	True Positive (TP)	False Negative (FN)
Negative	False Positive (FP)	True Negative (TN)



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The Confusion Matrix: Multi-class

Table 1

	Prediction		
Actual	a	b	c
a	T_a	F_{ab}	F_{ac}
b	F_{ba}	T_b	F_{bc}
c	F_{ca}	F_{cb}	T_c

Table 2

	Prediction		
Actual	TP	FP	FN
a	T_a	$F_{ba} + F_{ca}$	$F_{ab} + F_{ac}$
...	...	...	...

and N is the sum over all cells.



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Accuracy

$$\text{Accuracy} = \frac{(\text{TP}+\text{TN})}{(\text{TP}+\text{FP}+\text{FN}+\text{TN})}$$

or

$$\text{Accuracy} = \frac{T_a + T_b + T_c}{N}$$



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or

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Question: Can you see a problem with Accuracy?



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Precision

Of all the predicted positives, how many are actually positive?

$$\text{Precision} = \frac{(\text{TP})}{(\text{TP}+\text{FP})}$$

or

$$\text{Precision}_a = \frac{(T_a)}{T_a + F_{ba} + F_{ca}}$$

All predicted a: $T_a + F_{ba} + F_{ca}$

If we want one precision estimate for all classes:

1. Macro-average ($\text{Precision}_a, ..., \text{Precision}_c$)
2. Micro-average (use Table 2)



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Recall

Of all positives, how many are predicted correctly (recalled)?



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Recall

Of all positives, how many are predicted correctly (recalled)?

$$\text{Recall} = \frac{(\text{TP})}{(\text{TP}+\text{FN})}$$

and

$$\text{Recall}_a = \frac{(T_a)}{T_a + F_{ab} + F_{ac}}$$

All true/actual a: $T_a + F_{ab} + F_{ac}$

If we want one recall estimate for all classes:

1. Macro-average $(\text{Recall}_a, ..., \text{Recall}_c)$
2. Micro-average (use Table 2)



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Sensitivity and specificity

$$\text{Sensitivity} = \text{Recall of positive class} = \frac{\text{TP}}{\text{TP+FN}}$$

and

$$\text{Specificity} = \text{Recall of negative class} = \frac{\text{TN}}{\text{TN+FP}}$$



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F1-score

Harmonic mean of Precision and Recall.

$$F_1 = 2\frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

Question: What happens if Precision or Recall goes toward zero/one?



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Harmonic mean of Precision and Recall.

$$F_1 = 2\frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

Question: What happens if Precision or Recall goes toward zero/one? Very common performance measurement in practice.

If we want one F1 estimate for all classes:

1. Macro-average ($F_{1a}, ..., F_{1c}$)
2. Micro-average (use Table 2)



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Example

Say that we want to classify spam vs. ham.



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Example

Say that we want to classify spam vs. ham.

	$\hat{f}(x) = \text{ham}$	$\hat{f}(x) = \text{spam}$
$y = \text{ham}$	515	91
$y = \text{spam}$	85	569

The cell counts yield us estimates of

1. Accuracy: $\frac{515+569}{515+91+85+569} \approx 0.86$
2. Precision(ham): $\frac{515}{515+85} \approx 0.86$
3. Recall(ham): $\frac{515}{515+91} \approx 0.85$
4. F_1(ham): $\frac{2 \cdot 0.85 \cdot 0.86}{0.85+0.86} \approx 0.855$



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4. F_1(ham): $\frac{2 \cdot 0.85 \cdot 0.86}{0.85+0.86} \approx 0.855$

Let $\hat{\pi}_i$ be the predicted probability of spam. Predict spam when $\hat{\pi}_i > 0.5$, and ham otherwise.

What if we choose another cut-off level $\hat{\pi}_i > \alpha$ instead?



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Classification tables

$\alpha = 0.5$	$\hat{f}(x) = \text{ham}$	$\hat{f}(x) = \text{spam}$
$y = \text{ham}$	515	91
$y = \text{spam}$	85	569



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Classification tables

$\alpha = 0.5$	$\hat{f}(x) = \text{ham}$	$\hat{f}(x) = \text{spam}$
$y = \text{ham}$	515	91
$y = \text{spam}$	85	569

Now let $\alpha = 0.3$ instead, so that we are more prone to say that $\hat{y} = \text{spam}$:

$\alpha = 0.3$	$\hat{f}(x) = \text{ham}$	$\hat{f}(x) = \text{spam}$
$y = \text{ham}$	462	144
$y = \text{spam}$	38	616

The cell counts yield us estimates of

1. Accuracy: $\frac{462+616}{462+38+144+616} \approx 0.86$
2. Precision(ham): $\frac{462}{462+38} \approx 0.92$
3. Recall(ham): $\frac{462}{462+144} \approx 0.76$
4. F_1(ham): $\frac{2 \cdot 0.92 \cdot 0.76}{0.92+0.76} \approx 0.83$

For ham, precision has increased, but recall has decreased.



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Pair discussion: is high accuracy enough?

A dataset contains 1,001 legitimate emails (ham) and 17 spam emails. A classifier predicts ham for every email.

Pairs, 1–2 minutes:

1. Is this classifier useful for filtering spam? Explain.



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A more problematic example

A highly unbalanced example. 1001 ham and 17 spam.

Our new classifier: Everything is ham!



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A highly unbalanced example. 1001 ham and 17 spam.

Our new classifier: Everything is ham!

	$\hat{f}(x) = \text{ham}$	$\hat{f}(x) = \text{spam}$
$y = \text{ham}$	1001	0
$y = \text{spam}$	17	0



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	$\hat{f}(x) = \text{ham}$	$\hat{f}(x) = \text{spam}$
$y = \text{ham}$	1001	0
$y = \text{spam}$	17	0

The cell counts yield us estimates of

1. Accuracy: $\frac{1001}{1018} \approx 0.983$
2. Precision(ham): $\frac{1001}{1001+17} \approx 0.983$
3. Recall(ham): $\frac{1001}{1001+0} = 1$
4. F_1(ham): $\frac{2 \cdot 1001}{2 \cdot 1001+17} \approx 0.992$



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$y = \text{ham}$	1001	0
$y = \text{spam}$	17	0

The cell counts yield us estimates of

1. Accuracy: $\frac{1001}{1018} \approx 0.983$
2. Precision(ham): $\frac{1001}{1001+17} \approx 0.983$
3. Recall(ham): $\frac{1001}{1001+0} = 1$
4. F_1(ham): $\frac{2 \cdot 1001}{2 \cdot 1001+17} \approx 0.992$
5. Precision(spam) is undefined: no predicted spam.
6. F_1(spam): $\frac{2TP}{2TP+FP+FN} = \frac{0}{17} = 0$.
7. Taking ham as positive, specificity is $0/17 = 0$.



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Section 3

Test and training error



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Test Error

- The main error of interest - generalization error
- Conditional Test Error
  (Performance for the model trained on actual training data):

$$\mathrm{Err}_{\mathcal{T}} = \mathbb{E}_{X_0, Y_0}\left[L(Y_0, \hat{f}_{\mathcal{T}}(X_0)) \mid \mathcal{T}\right]$$

- Expected Test Error
  (Model performance over different training datasets):

$$\mathrm{Err} = \mathbb{E}_{\mathcal{T}}[\mathrm{Err}_{\mathcal{T}}]$$

- The outer expectation averages over training datasets of the same size, for a fixed learning procedure.



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Training Error

- Training error: The loss the algorithm tries to minimize
- The Error in the training data:

$$\overline{\text{err}} = \frac{1}{N} \sum_{i=1}^{N} L(y_i, \hat{f}(x_i))$$

where $L(y, x)$ is the loss function.



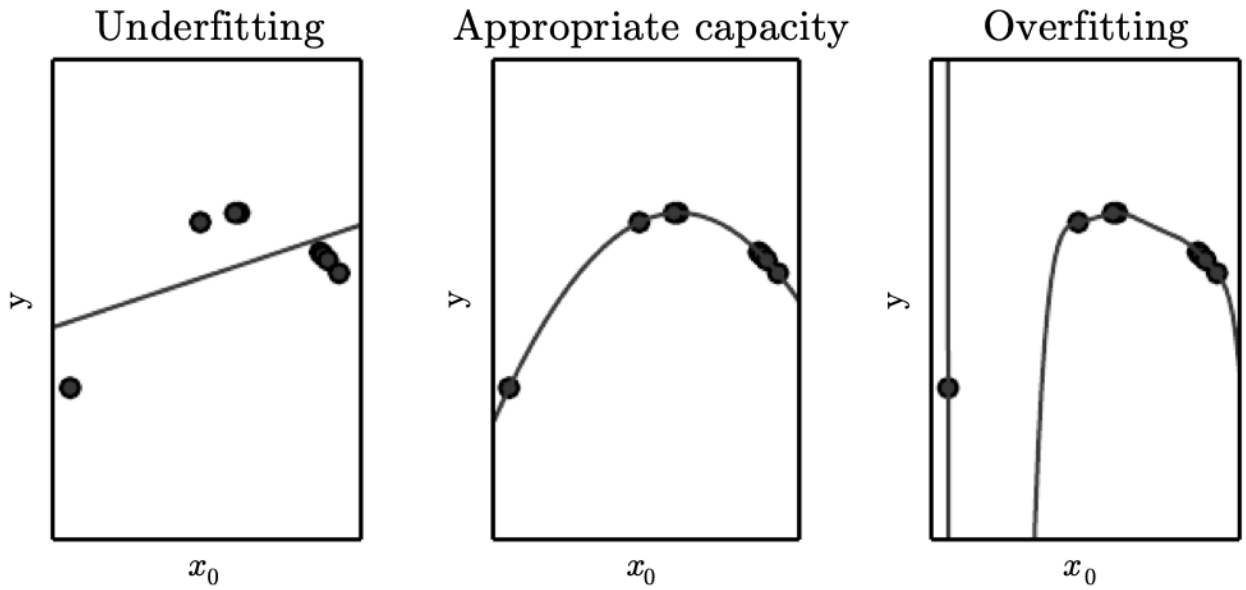
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Model complexity/capacity

- Model complexity/capacity: The flexibility of the model.
- Underfitting: Too low capacity of model
- Overfitting: Too high capacity of model
- Example: Polynomial regression with higher order terms

Figure: Model complexity (Goodfellow et al, 2017, Figure 5.2)



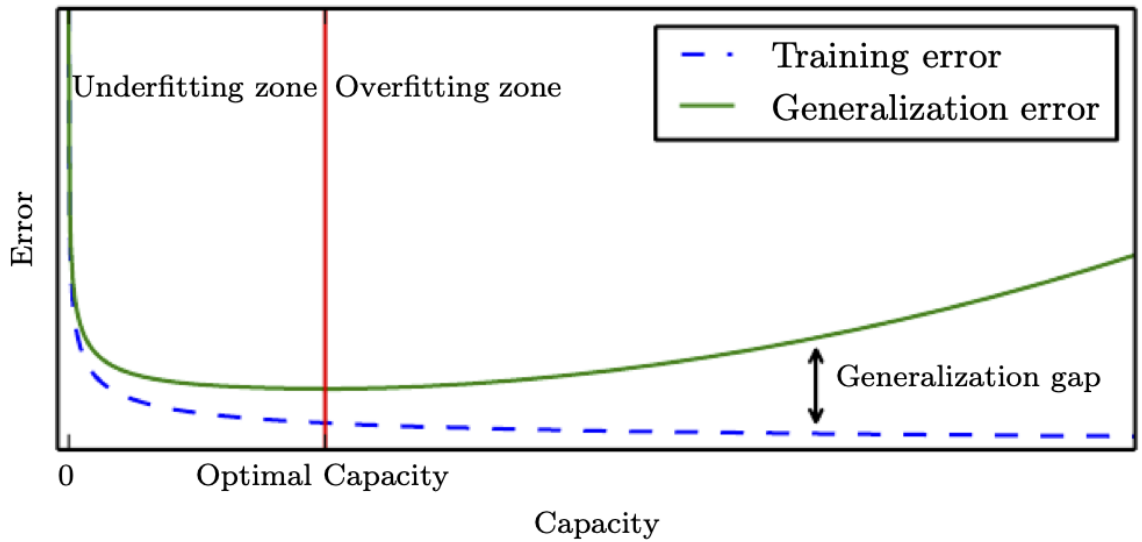


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Training, test, and complexity

Figure: Test, training, and model complexity (Goodfellow et al, 2017, Figure 5.3)





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How to estimate the Test Error (Model Assessment)

- We set aside a test set from the data
- Use as the last step to estimate the generalization error
- Keep untouched until final assessment; do not adapt model choices to its results.



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How to estimate the Test Error (Model Assessment)

- We set aside a test set from the data
- Use as the last step to estimate the generalization error
- Keep untouched until final assessment; do not adapt model choices to its results.
- Size of testset:
  - Common suggestion is 10%, but
  - It is a statistical estimation problem (choice of sampling size)



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Pair discussion: choosing the best test score

We evaluate 20 models on the same test set. Their true generalisation errors are identical, but their measured test errors differ because the test set is finite.

Pairs, 1–2 minutes:

1. We choose the model with the lowest measured test error. Would you trust that score as an estimate of its future error? Why?
2. How could we select a model and still obtain an independent final assessment?



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Model selection and final assessment

- Selecting the lowest validation error also selects favourable evaluation noise: the winning score is optimistic.
- Use validation data or CV to choose models and hyperparameters.
- Assess the selected procedure on an untouched test set.



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Section 5

Bias and Variance



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Bias and Variance

Assume $Y_0 = f(X_0) + \epsilon_0$, with $\mathbb{E}[\epsilon_0 \mid X_0] = 0$ and $V(\epsilon_0 \mid X_0) = \sigma_\epsilon^2$.

The test observation is independent of training data $\mathcal{T}$. At a fixed input x_0, averaging over training data and new test noise:

$$\begin{aligned}\mathrm{Err}(x_0) &= \mathbb{E}_{\mathcal{T},Y_0}\left[(Y_0 - \hat{f}_{\mathcal{T}}(x_0))^2 \mid X_0 = x_0\right] \\ &= \sigma_\epsilon^2 + \underbrace{\left(\mathbb{E}_{\mathcal{T}}[\hat{f}_{\mathcal{T}}(x_0)] - f(x_0)\right)^2}_{\text{bias}^2} + \underbrace{V_{\mathcal{T}}(\hat{f}_{\mathcal{T}}(x_0))}_{\text{variance}}.\end{aligned}$$



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- *Bias*: How close can $\hat{f}$ get to the true model f
- *Variance*: The variability of the predictions from $\hat{f}$
- *Irreducible/Bayes error*: The minimum possible error



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Bias and Variance: Linear regression

For ordinary least squares, $\hat{f}(x_i) = x_i^T \hat{\beta}$. Assume a fixed, full-rank design X, independent errors with zero mean and variance σ_ϵ^2.

Averaging test error at the *training inputs* (with fresh outcomes):

$$\frac{1}{N}\sum_{i=1}^{N} \mathrm{Err}(x_i) = \underbrace{\sigma_\epsilon^2}_{\text{irreducible}} + \underbrace{\frac{1}{N}\sum_{i=1}^{N}\left(f(x_i) - \mathbb{E}[\hat{f}(x_i)]\right)^2}_{\text{bias}^2} + \underbrace{\frac{p}{N}\sigma_\epsilon^2}_{\text{variance}}.$$

Here p counts fitted coefficients, including the intercept. Expectations are over new training responses at fixed X.

At a new input x_0, prediction variance is instead $\sigma_\epsilon^2 x_0^T (X^T X)^{-1} x_0$.

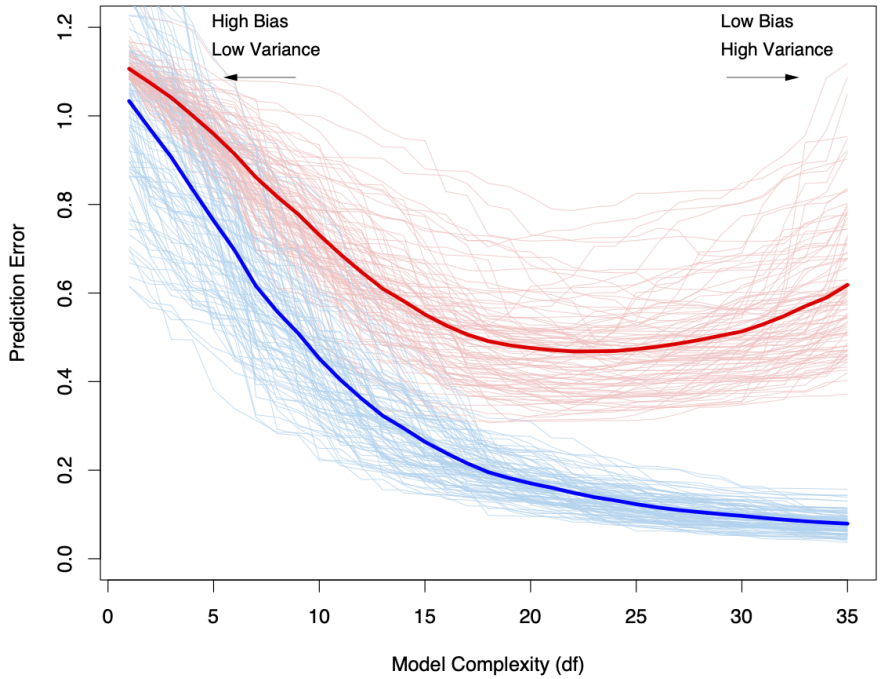


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Bias and Variance

Figure: Test (red), training (blue), and model complexity (Hastie et al, 2009, Figure 7)



- High Bias: Underfit
- High Variance: Overfit
- High irreducible error: limits achievable prediction accuracy

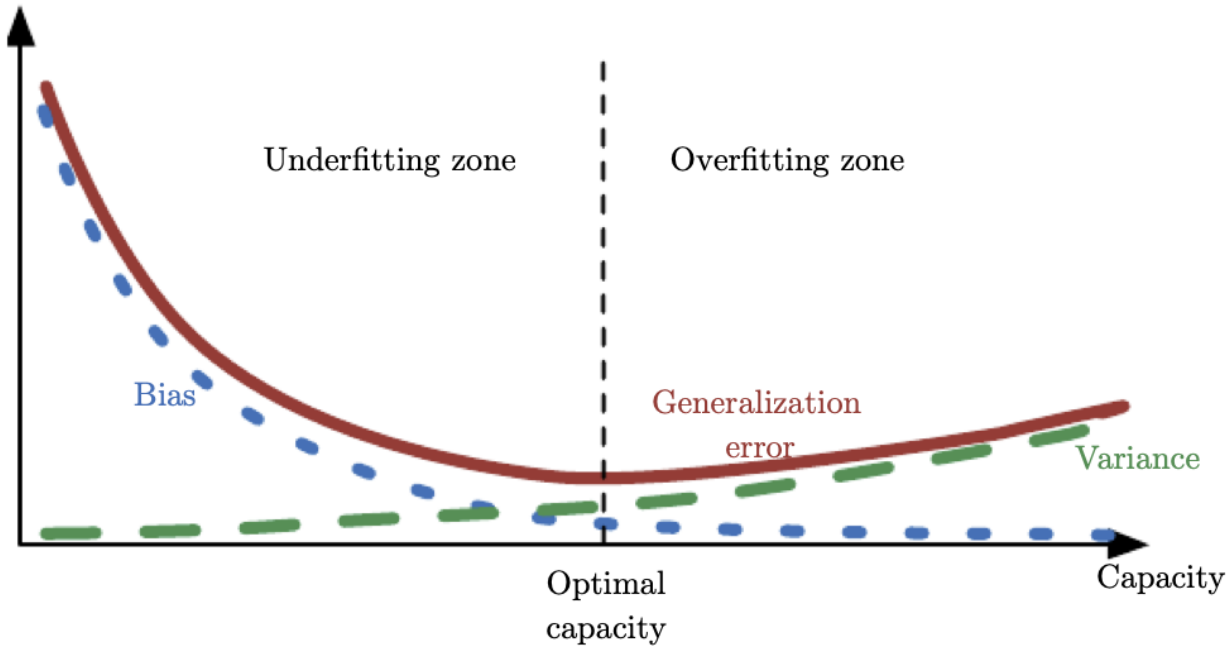


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Bias and Variance

Figure: Bias and variance (Goodfellow et al., 2017, Figure 5.6)



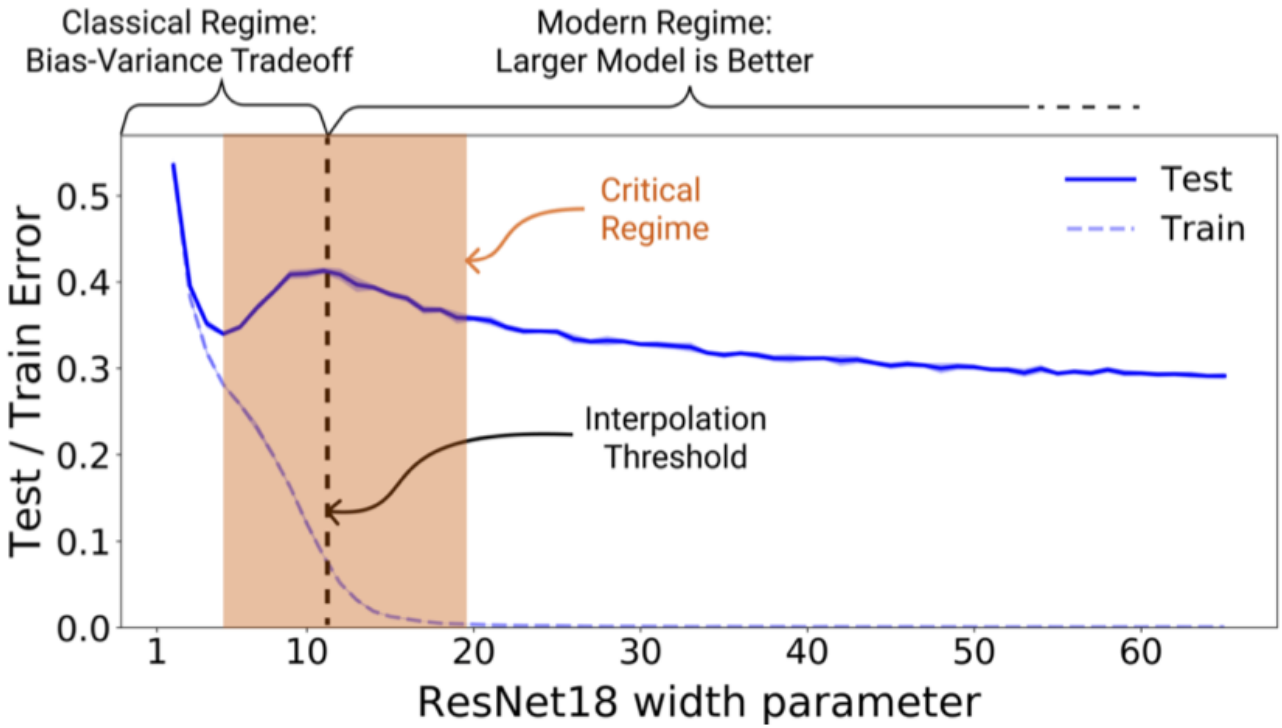


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The double descent of large models

Figure: The double descent of large models (Nakkiran et al., 2019)





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Questions?

Questions?



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Section 6

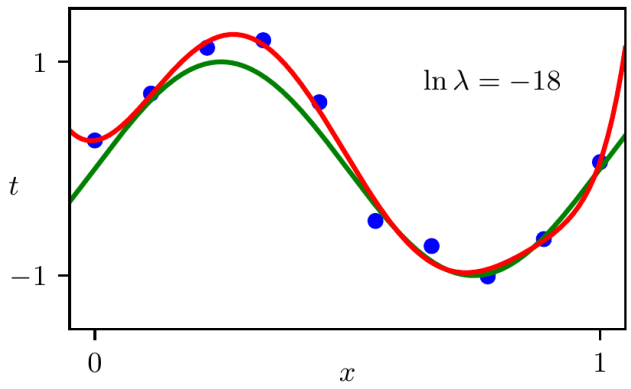
Regularisation

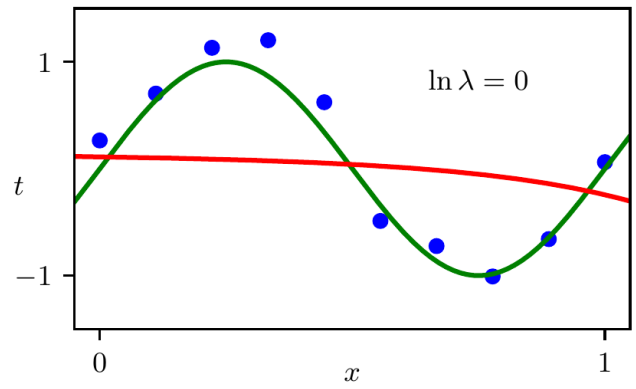


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Why regularise?





Same degree-9 polynomial, two penalty strengths. Stronger regularisation can underfit. Bishop & Bishop (2024), Fig. 1.9.

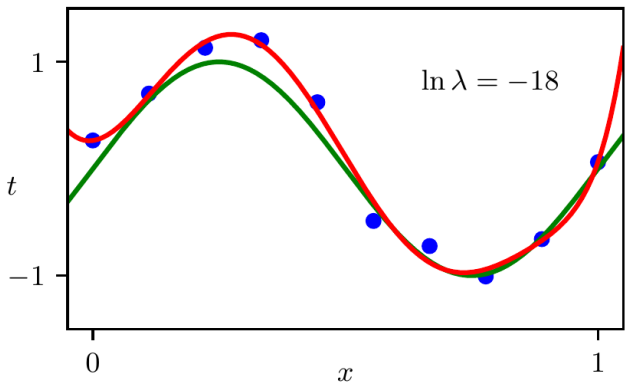
- Many or highly correlated predictors can give unstable or non-unique coefficient estimates.

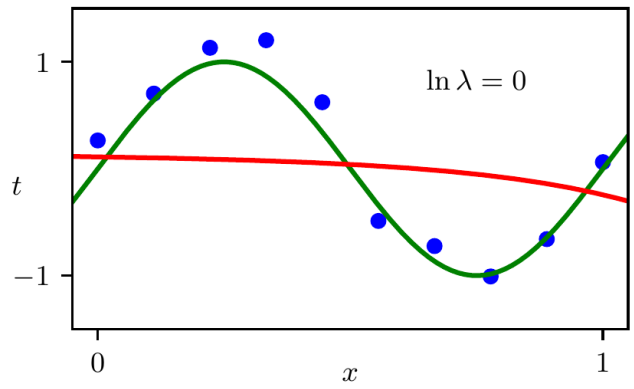


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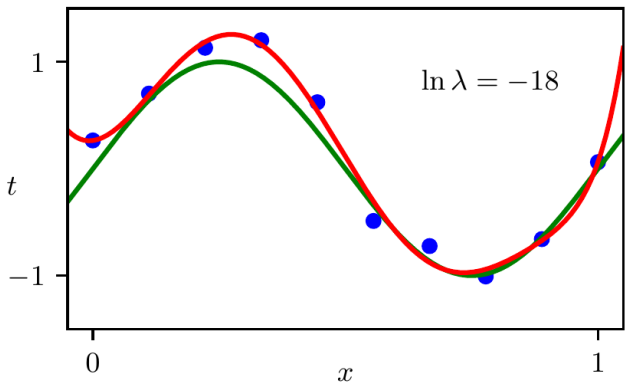
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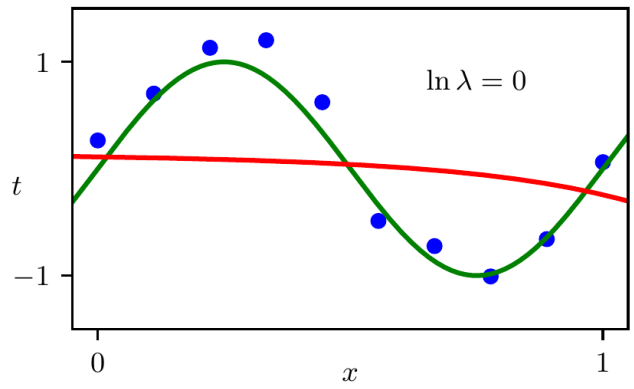


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- Many or highly correlated predictors can give unstable or non-unique coefficient estimates.
- Different coefficients may fit the training data equally well but predict differently on new observations.
- Regularisation favours some solutions over others to improve generalisation.



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Regularisation controls effective complexity

$$\hat{\theta}_\lambda = \arg\min_\theta \left\{ \frac{1}{N} \sum_{i=1}^{N} \ell_i(\theta) + \lambda \Omega(\theta) \right\}, \qquad \lambda \geq 0.$$

- θ is fitted to training data; λ is a hyperparameter.



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- Stronger penalties typically reduce variance at the cost of bias.



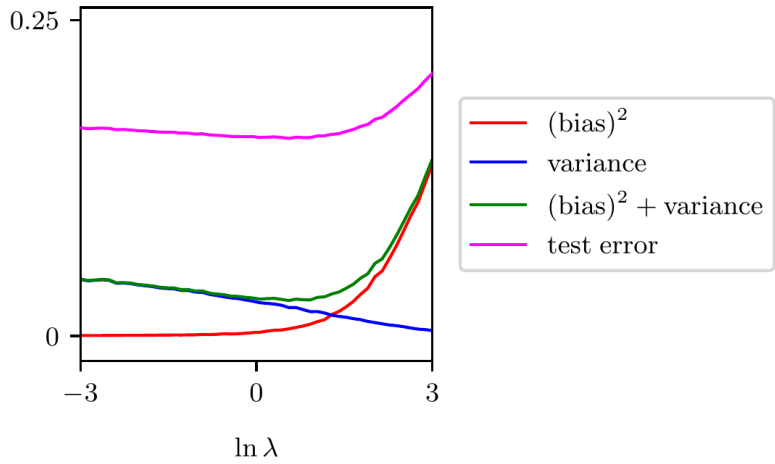
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- θ is fitted to training data; λ is a hyperparameter.
- Stronger penalties typically reduce variance at the cost of bias.
- Effective flexibility changes without removing available predictors. Too much regularisation can underfit.



Bishop & Bishop (2024), Fig. 4.8



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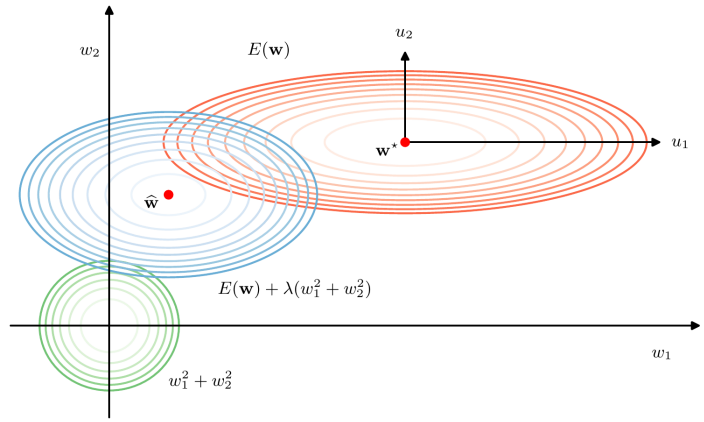
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Ridge regression

For $\hat{y}_i = \beta_0 + \mathbf{x}_i^T \beta$, minimise

$$\frac{1}{2N} \sum_{i=1}^{N} (y_i - \beta_0 - \mathbf{x}_i^T \beta)^2 + \frac{\lambda}{2} \sum_{j=1}^{p} \beta_j^2.$$

- Shrinks coefficients close to zero.
- The intercept β_0 is not penalised.



Bishop & Bishop (2024), Fig. 9.3



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Lasso regression

For the same linear prediction model, minimise

$$\frac{1}{2N}\sum_{i=1}^{N}(y_i - \beta_0 - \mathbf{x}_i^T\beta)^2 + \lambda\sum_{j=1}^{p}|\beta_j|.$$

- An absolute-value (L_1) penalty can set coefficients exactly to zero.



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- An absolute-value (L_1) penalty can set coefficients exactly to zero.
- This gives a sparse model: some predictors are excluded.
- Correlated predictors may lead to unstable variable selection.
- A selected predictor is not necessarily a causal or "true" variable.
- As with ridge, leave the intercept unpenalised.



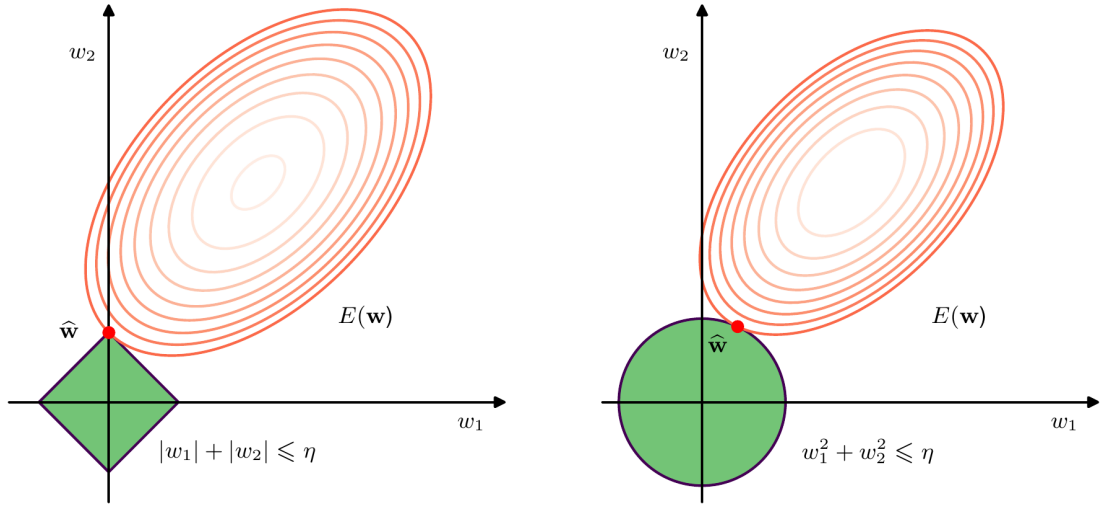
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Why does lasso produce zeros?

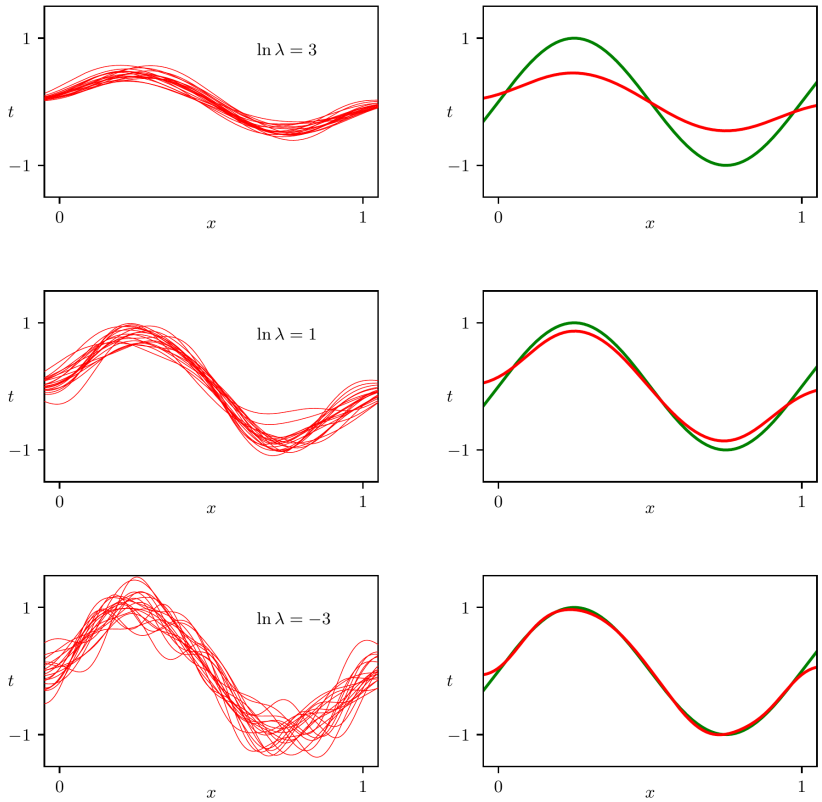
Equivalent constrained views, for suitable bounds c:

Lasso: $\sum_j |\beta_j| \leq c$, Ridge: $\sum_j \beta_j^2 \leq c$.



Bishop & Bishop (2024), Fig. 9.6

What does λ do?



Bishop & Bishop (2024), Fig. 4.7

- $\lambda = 0$: no penalty. Larger λ: stronger shrinkage.



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Pair discussion: ridge or lasso?

Pairs, 1–2 minutes: Would you favour ridge or lasso if many predictors have small effects? What if only a few are useful?



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Scaling and the intercept

- Changing a predictor from metres to centimetres divides its coefficient by 100, changing its penalty despite identical predictions.
- Standardisation puts predictors on comparable scales:

$$z_{ij} = \frac{x_{ij} - \bar{x}_j}{s_j}.$$

- Usually do not penalise the intercept. It sets the overall level, rather than the effects of predictors.
- Penalty strength depends on loss scaling: a sum and a mean require different numerical λ values.



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A probabilistic interpretation: MAP

$$\hat{\theta}_{\mathrm{MAP}} = \arg\min_{\theta} \left\{ -\log p(\mathbf{y} \mid X, \theta) - \log p(\theta) \right\}.$$



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- Independent zero-mean Gaussian priors give quadratic penalties (ridge).



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- Stronger concentration around zero means stronger shrinkage.



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Logistic ridge and gradient descent

With intercept first, $\theta = (\theta_1, \ldots, \theta_P)^T$:

$$L_\lambda(\theta) = \underbrace{\frac{1}{N}\sum_i \ell_i(\theta)}_{\text{mean negative log-likelihood}} + \frac{\lambda}{2}\sum_{j=2}^{P}\theta_j^2.$$

$$\nabla L_\lambda(\theta) = \nabla L(\theta) + \lambda(0, \theta_2, \ldots, \theta_P)^T.$$

- For mini-batches, estimate the data gradient. Add the exact penalty gradient at every update.



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- For mini-batches, estimate the data gradient. Add the exact penalty gradient at every update.
- The intercept receives no shrinkage term.
- This is Assignment 1's optional VG logistic ridge task.



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Choosing λ in practice

- In `glmnet`, `alpha = 0` selects ridge and `alpha = 1` selects lasso; `lambda` controls penalty strength.
- Compare candidate λ values using the same validation folds.
- Assignment 1 asks for the mean of ten fold RMSEs.

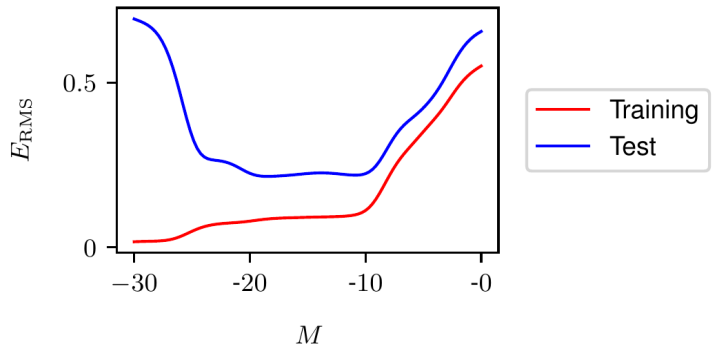


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- Compare candidate λ values using the same validation folds.
- Assignment 1 asks for the mean of ten fold RMSEs.
- Refit with the selected λ on all training data. Assess once on the untouched test set.



Bishop & Bishop (2024), Fig. 1.10. Horizontal axis should read ln λ.

How do we choose λ without using the test set?



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Section 7

Cross-validation



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Cross-Validation

- We want to estimate the generalization error Err for different models and to choose the best model where

$$\begin{aligned} \text{Err} &= \mathbb{E}_{p(X,Y)}(\text{Err}_{\mathcal{T}}) \\ &= \mathbb{E}_{p(X,Y)}(\mathbb{E}_{p(X_0,Y_0)}(L(Y_0, X_0)|\mathcal{T})) \end{aligned}$$

- Cross-Validation is probably the most popular approach to estimate Err and choose between models because it is:
  1. Conceptually easy to understand
  2. Easy to implement
  3. No need for rules-of-thumbs to verify that it is applicable
  4. Applicable to many different types of models
  5. Flexible for the use case at hand



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  4. Applicable to many different types of models
  5. Flexible for the use case at hand
- Common approach to choose hyperparameters (that is a model choice)



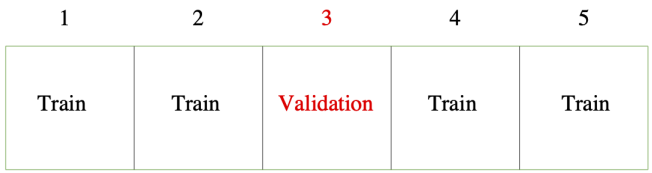
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The Cross-Validation Algorithm

Figure: Cross-Validation (Hastie et al, 2009, p. 222, 242)





1. Split data in K folds
2. For each fold $k = 1, 2, ..., K$
   - 2.1 Fit $\hat{f}^{(-k)}$ using all observations outside fold k
   - 2.2 Predict observations in fold k; fit preprocessing and feature selection using training folds only

$$\widehat{\text{Err}}_{CV} = \frac{1}{N} \sum_{i=1}^{N} L(y_i, \hat{f}^{(-k(i))}(x_i))$$

where $k(i)$ is the fold containing observation i. Any tuning assessed by a held-out fold must use only its training data.



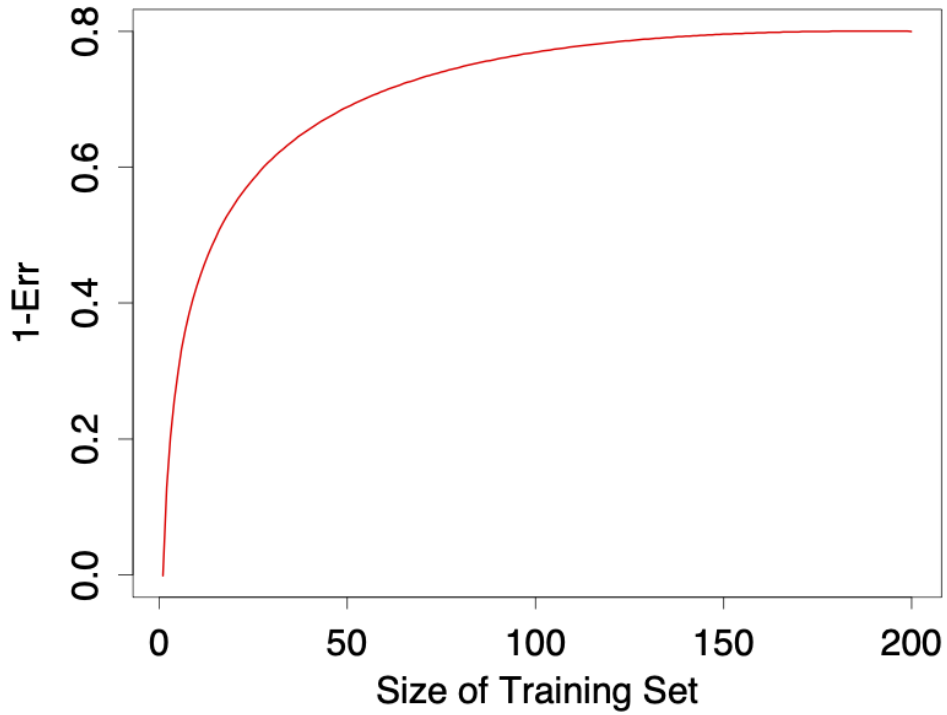
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The Bias of Cross-Validation

- Cross-validation estimation of Err will be biased
- The training data size is smaller than the full data

Figure: Cross-Validation Bias (Hastie et al, 2009, Fig. 7.8)





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K-fold Cross Validation

- Common choices are $K \in \{2, 5, 10\}$
- Smaller K often increases training-size bias; this is not universal
- Larger K is computationally more costly
- $K = 10$ is a common approach



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Leave-One-Out Cross Validation

- When $K = N$
- Benefits
  - Often small training-size bias: each fit uses $N - 1$ observations
  - Sometimes we only need to train our model once
- Drawbacks
  - Higher Variance in estimate of Err
  - Can be computationally costly (naive implementation)
  - Can be unstable/less robust in some settings



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The role of the data generating process

- we assume that testset and train set are different observations from the same data generating process

$$\mathbf{d} = \{(y_i, \mathbf{x}_i), i = 1, ..., n\} \sim P_{y,x}$$

- The (naive) assumption: independence
- Things that can go wrong:
  - temporal leakage or concept drift
  - duplicated observations
  - unrepresentative sampling or distribution shift



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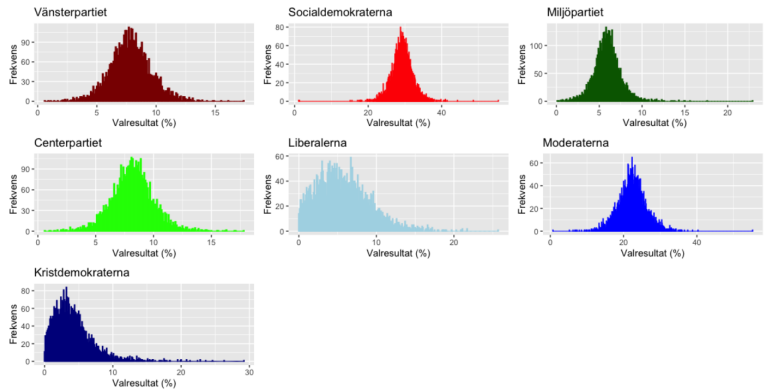
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Example: Election prediction

- We want to predict the next election
- The relationship may change over time (concept drift)
- Example in Frölander and Uddhammar (2021) and Olsson and Ölfvingsson (2021)
  1. LOO-CV on the elections 1973-2014
  2. The 2018 election held out for later assessment
- LOO may train on later elections. For forecasting, use forward/rolling splits: train only on earlier elections.

Figure: Predictive distr. (Olsson and Ölfvingsson, 2021, Fig. 6)





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What is CV estimating?

$$\text{Err} = \mathbb{E}_{p(X,Y)}(\text{Err}_{\mathcal{T}})$$

1. What do we want CV to estimate, Err or $\text{Err}_{\mathcal{T}}$?



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What is CV estimating?

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2. What is CV estimating?



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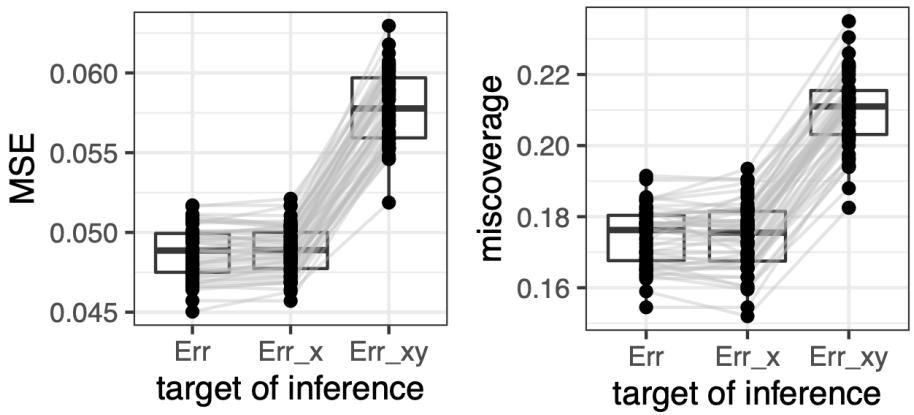
What is CV estimating?

Further, assume the true model is

$$y_i = \mathbf{x}_i^T \theta + \epsilon_i$$

where $\epsilon_i \sim N(0, \sigma^2)$.

Figure: MSE and miscoverage with $\text{Err}_{XY} = \text{Err}_{\mathcal{T}}$ (Bates et al., arXiv:2104.00673)





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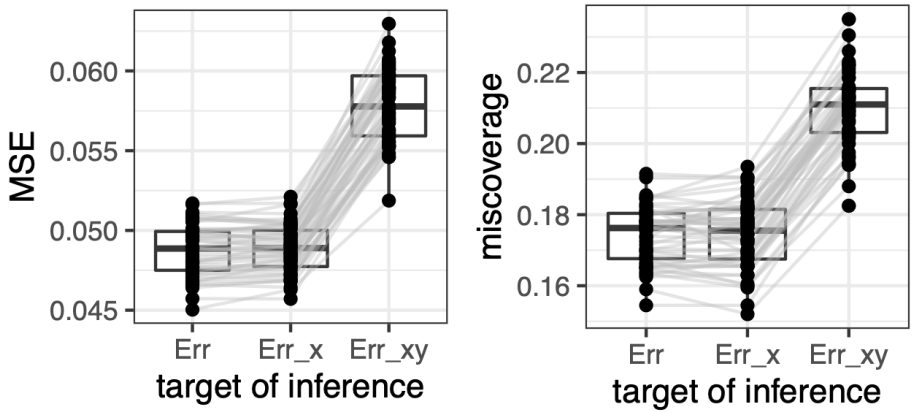
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Two take-aways:

1. CV is estimating Err (see Bates et al, 2022)



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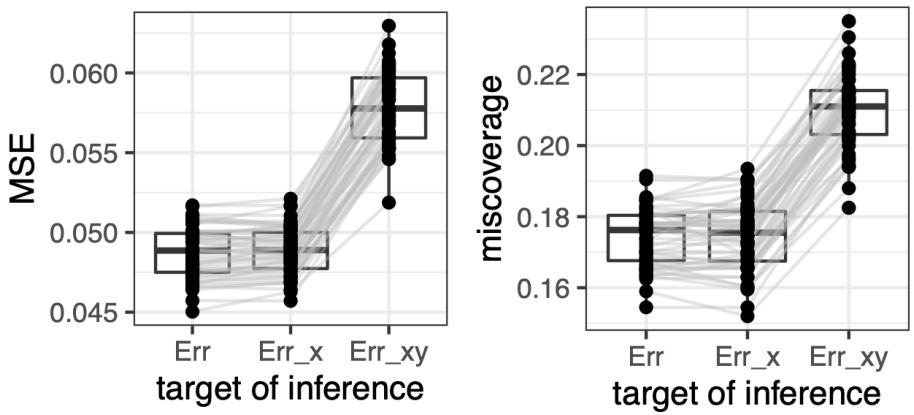
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Two take-aways:

1. CV is estimating Err (see Bates et al, 2022)
2. Fold errors are dependent: naive SEs can underestimate uncertainty (Bengio and Grandvalet, 2004).



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Test error and (cross-)validation error?

1. What is the difference between the validation and test error?



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Test error and (cross-)validation error?

1. What is the difference between the validation and test error?
2. Why use cross-validation instead of holding out one validation fold?



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Questions?

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